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Subject :- .Net Programming

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Practical:01

Aim: Accept a character from console and check the case of the character.

Program.cs:-

```
using System;
class Program
{
    static void Main()
    {
        Console.Write("Enter a character: ");
        char ch = Console.ReadKey().KeyChar;
        Console.WriteLine();
        if (char.IsUpper(ch)){
            Console.WriteLine($"{ch} is an UPPERCASE letter.");
        }
        else if (char.IsLower(ch)){
            Console.WriteLine($"{ch} is a LOWERCASE letter.");
        }
        else if (char.IsDigit(ch)){
            Console.WriteLine($"{ch} is a DIGIT.");
        }
        else{
            Console.WriteLine($"{ch} is a SPECIAL CHARACTER.");
        }
        Console.WriteLine("Press any key to exit...");
        Console.ReadKey();
    }
}
```

Output:-

```
PS C:\Users\AmanDiktiya\Desktop\dotnet\Practicals> dotnet run
Enter a character: A
A is an UPPERCASE letter.
Press any key to exit...
PS C:\Users\AmanDiktiya\Desktop\dotnet\Practicals> dotnet run
Enter a character: 8
8 is a DIGIT.
Press any key to exit...
PS C:\Users\AmanDiktiya\Desktop\dotnet\Practicals>
```

Practical:02

Aim: Write a program to accept any character from keyboard and display whether it is vowel or not.

Program.cs:-

```
using System;
class Program
{
    static void Main()
    {
        Console.Write("Enter a character: ");
        char ch = Char.ToLower(Console.ReadKey().KeyChar);
        Console.WriteLine();

        if (ch == 'a' || ch == 'e' || ch == 'i' || ch == 'o' || ch == 'u')
        {
            Console.WriteLine("The character is a vowel.");
        }
        else
        {
            Console.WriteLine("The character is not a vowel.");
        }
    }
}
```

Output:-

```
PS C:\Users\AmanDiktiya\Desktop\dotnet\Practicals> scriptcs "c:\Users\AmanDiktiya\Desktop\dotnet\Practicals\Practical1.cs"
PS C:\Users\AmanDiktiya\Desktop\dotnet\Practicals> dotnet run
Enter a character: a
The character is a vowel.
PS C:\Users\AmanDiktiya\Desktop\dotnet\Practicals> dotnet run
Enter a character: b
The character is not a vowel.
PS C:\Users\AmanDiktiya\Desktop\dotnet\Practicals>
```

Practical:03

AIM : Write a program to accept a string and convert the case of the characters.

Program.cs:-

```
using System;
class Program
{
    static void Main()
    {
        // Accepting input string from the user
        Console.Write("Enter a string: ");
        string inputString = Console.ReadLine();

        // Convert case in three ways
        string uppercaseString = inputString.ToUpper();
        string lowercaseString = inputString.ToLower();
        string swapcaseString = SwapCase(inputString);

        // Display results
        Console.WriteLine("\nConverted Strings:");
        Console.WriteLine("1. Uppercase: " + uppercaseString);
        Console.WriteLine("2. Lowercase: " + lowercaseString);
        Console.WriteLine("3. Swap Case: " + swapcaseString);
    }

    static string SwapCase(string str)
    {
        char[] charArray = str.ToCharArray();
        for (int i = 0; i < charArray.Length; i++)
        {
            if (char.IsUpper(charArray[i]))
                charArray[i] = char.ToLower(charArray[i]);
            else if (char.IsLower(charArray[i]))
                charArray[i] = char.ToUpper(charArray[i]);
        }
        return new string(charArray);
    }
}
```

OUTPUT :

```
PS C:\Users\AmanDiktiya\Desktop\dotnet> scriptcs "c:\Users\AmanDiktiya\Desktop\dotnet\Practicals\Program.cs"
PS C:\Users\AmanDiktiya\Desktop\dotnet> cd Practicals
PS C:\Users\AmanDiktiya\Desktop\dotnet\Practicals> dotnet run
Enter a string: aman diktiya

Converted Strings:
1. Uppercase: AMAN DIKTIYA
2. Lowercase: aman diktiya
3. Swap Case: AMAN DIKTIYA
PS C:\Users\AmanDiktiya\Desktop\dotnet\Practicals>
```

Practical:04

Aim - Develop a menu based application to implement a text editor with cut, copy, paste, save and close operation.

Program.cs:-

```
using System;
using System.Collections.Generic;
class TextEditor
{
   private string text = "";
   private string clipboard = "";
   public void Run()
   {
      while (true)
      {
         Console.WriteLine("\nText Editor Menu:");
         Console.WriteLine("1. Enter Text");
         Console.WriteLine("2. Cut");
         Console.WriteLine("3. Copy");
         Console.WriteLine("4. Paste");
         Console.WriteLine("5. Save");
         Console.WriteLine("6. Close");
         Console.Write("Choose an option: ");
        string choice = Console.ReadLine();
         switch (choice)
         {
            case "1":
               Console.Write("Enter text: ");
               text = Console.ReadLine();
               break;
            case "2":
               Cut();
               break;
            case "3":
               Copy();
               break;
            case "4":
               Paste();
               break;
            case "5":
```

```
                Save();
                break;
            case "6":
                Console.WriteLine("Closing Editor...");
                return;
            default:
                Console.WriteLine("Invalid choice. Try again.");
                break;
        }
    }
}
private void Cut()
{
    Console.Write("Enter text to cut: ");
    string input = Console.ReadLine();
    if (text.Contains(input))
    {
        clipboard = input;
        text = text.Replace(input, "");
        Console.WriteLine("Text cut successfully.");
    }
    else
    {
        Console.WriteLine("Text not found.");
    }
}

private void Copy()
{
    Console.Write("Enter text to copy: ");
    string input = Console.ReadLine();
    if (text.Contains(input))
    {
        clipboard = input;
        Console.WriteLine("Text copied successfully.");
    }
    else
    {
        Console.WriteLine("Text not found.");
    }
}
private void Paste()
{
    text += clipboard;
```

```
            Console.WriteLine("Pasted successfully.");
        }

        private void Save()
        {
            Console.WriteLine("Saving text...");
            Console.WriteLine("Saved Text: " + text);
        }
        public static void Main()
        {
            TextEditor editor = new TextEditor();
            editor.Run();
        }
    }
```

Output :

```
Text Editor Menu:
1. Enter Text
2. Cut
3. Copy
4. Paste
5. Save
6. Close
Choose an option: 1
Enter text: Aman Diktiya
```

```
Text Editor Menu:
1. Enter Text
2. Cut
3. Copy
4. Paste
5. Save
6. Close
Choose an option: 3
Enter text to copy: Aman
Text copied successfully.
```

```
Text Editor Menu:
1. Enter Text
2. Cut
3. Copy
5. Save
6. Close
Choose an option: 4
Pasted successfully.
```

```
Text Editor Menu:
1. Enter Text
2. Cut
3. Copy
4. Paste
5. Save
6. Close
Choose an option: 6
Closing Editor...
```

Practical:05

Aim - Write a program to implement a calculator with memory and recall operations.

Program.cs:-

```
using System;
class Calculator
{
   private double memory = 0;
   public void Run()
   {
      while (true)
      {
         Console.WriteLine("\nCalculator Menu:");
         Console.WriteLine("1. Add");
         Console.WriteLine("2. Subtract");
         Console.WriteLine("3. Multiply");
         Console.WriteLine("4. Divide");
         Console.WriteLine("5. Memory Store");
         Console.WriteLine("6. Memory Recall");
         Console.WriteLine("7. Clear Memory");
         Console.WriteLine("8. Exit");
         Console.Write("Choose an option: ");
         string choice = Console.ReadLine();
         if (choice == "8")
         {
            Console.WriteLine("Exiting Calculator...");
            break;
         }
        double num1, num2;
         switch (choice)
         {
            case "1":
               (num1, num2) = GetTwoNumbers();
               Console.WriteLine("Result: " + (num1 + num2));
               break;
            case "2":
               (num1, num2) = GetTwoNumbers();
               Console.WriteLine("Result: " + (num1 - num2));
               break;
```

```
            case "3":
                (num1, num2) = GetTwoNumbers();
                Console.WriteLine("Result: " + (num1 * num2));
                break;
            case "4":
                (num1, num2) = GetTwoNumbers();
                if (num2 != 0)
                    Console.WriteLine("Result: " + (num1 / num2));
                else
                    Console.WriteLine("Error: Division by zero");
                break;
            case "5":
                Console.Write("Enter number to store in memory: ");
                memory = double.Parse(Console.ReadLine());
                Console.WriteLine("Number stored in memory.");
                break;
            case "6":
                Console.WriteLine("Memory Recall: " + memory);
                break;
            case "7":
                memory = 0;
                Console.WriteLine("Memory Cleared.");
                break;
            default:
                Console.WriteLine("Invalid choice. Try again.");
                break;
        }
    }
}

private (double, double) GetTwoNumbers()
{
    Console.Write("Enter first number: ");
    double num1 = double.Parse(Console.ReadLine());
    Console.Write("Enter second number: ");
    double num2 = double.Parse(Console.ReadLine());
    return (num1, num2);
}
public static void Main()
{
    Calculator calc = new Calculator();
    calc.Run();
}
}
```

Output :

```
Calculator Menu:
1. Add
2. Subtract
3. Multiply
4. Divide
5. Memory Store
6. Memory Recall
7. Clear Memory
8. Exit
Choose an option: 1
Enter first number: 26
Enter second number: 48
Result: 74
```

```
Calculator Menu:
1. Add
2. Subtract
3. Multiply
4. Divide
5. Memory Store
6. Memory Recall
7. Clear Memory
8. Exit
Choose an option: 2
Enter first number: 45
Enter second number: 20
Result: 25
```

```
Calculator Menu:
1. Add
2. Subtract
3. Multiply
4. Divide
5. Memory Store
6. Memory Recall
7. Clear Memory
8. Exit
Choose an option: 3
Enter first number: 12
Enter second number: 40
Result: 480
```

```
Calculator Menu:
1. Add
2. Subtract
3. Multiply
4. Divide
5. Memory Store
6. Memory Recall
7. Clear Memory
8. Exit
Choose an option: 5
Enter number to store in memory: 12
Number stored in memory.
```

```
Calculator Menu:
1. Add
2. Subtract
3. Multiply
4. Divide
5. Memory Store
6. Memory Recall
7. Clear Memory
8. Exit
Choose an option: 6
Memory Recall: 12
```

Practical:06

Aim - Develop a form in to pick a date from Calendar control and display the day, month, year details in separate text boxes.

Program.cs:-

```
using System;
using System.Globalization;
class Program
{
   static void Main()
   {
      Console.WriteLine("Select a date from the calendar below:");
      DateTime selectedDate = DateTime.Today;
      Console.WriteLine("Today's Date: " + selectedDate.ToString("dd-MM-yyyy"));
      Console.Write("Enter a future date (dd-MM-yyyy): ");
      string input = Console.ReadLine();
      if (DateTime.TryParseExact(input, "dd-MM-yyyy", null, DateTimeStyles.None,
out selectedDate))
      {
         Console.WriteLine("Selected Date Details:");
         Console.WriteLine("Day: " + selectedDate.Day);
         Console.WriteLine("Month: " + selectedDate.Month);
         Console.WriteLine("Year: " + selectedDate.Year);
      }
      else
      {
         Console.WriteLine("Invalid date format. Please enter in dd-MM-yyyy
format.");
      }
   }
}
```

Output :

```
PS C:\Users\AmanDiktiya\Desktop\dotnet\Practicals> dotnet run
Select a date from the calendar below:
Today's Date: 21-03-2026
Enter a future date (dd-MM-yyyy): 30-03-2026
Selected Date Details:
Day: 30
Month: 3
Year: 2026
PS C:\Users\AmanDiktiya\Desktop\dotnet\Practicals>
```

Practical:07

Aim - Develop a application to perform timer based quiz of 10 questions.

Program.cs:-

```
using System;
using System.Collections.Generic;
using System.Threading;
class Program
{
   static int score = 0;
   static int timeLimit = 10; // Time limit per question in seconds
   static void Main()
   {
      List<(string Question, string[] Options, int CorrectOption)> quiz = new
List<(string, string[], int)>
      {
         ("What is the capital of France?", new string[] {"1. Paris", "2. London", "3.
Berlin", "4. Madrid"}, 1),
         ("Which is the largest planet in our solar system?", new string[] {"1. Earth",
"2. Mars", "3. Jupiter", "4. Saturn"}, 3),
         ("Who wrote 'Hamlet'?", new string[] {"1. Dickens", "2. Shakespeare", "3.
Tolstoy", "4. Hemingway"}, 2),
         ("Which gas do plants use for photosynthesis?", new string[] {"1. Oxygen",
"2. Nitrogen", "3. Carbon Dioxide", "4. Hydrogen"}, 3),
         ("What is 5 + 7?", new string[] {"1. 10", "2. 12", "3. 14", "4. 15"}, 2),
         ("Which element has the atomic number 1?", new string[] {"1. Oxygen", "2.
Hydrogen", "3. Helium", "4. Carbon"}, 2),
         ("What is the currency of Japan?", new string[] {"1. Yen", "2. Dollar", "3.
Peso", "4. Rupee"}, 1),
         ("Who discovered gravity?", new string[] {"1. Einstein", "2. Newton", "3.
Galileo", "4. Kepler"}, 2),
         ("What is the square root of 64?", new string[] {"1. 6", "2. 7", "3. 8", "4. 9"},
3),
         ("Which ocean is the largest?", new string[] {"1. Atlantic", "2. Indian", "3.
Arctic", "4. Pacific"}, 4)
      };
      Console.WriteLine("Welcome to the Timer-based Quiz!\n");
      for (int i = 0; i < quiz.Count; i++)
      {
```

```
            Console.WriteLine($"Question {i + 1}: {quiz[i].Question}");
            foreach (var option in quiz[i].Options)
            {
                Console.WriteLine(option);
            }
            Console.Write("Enter your answer (1-4) within " + timeLimit + " seconds: ");
            string userInput = ReadInputWithTimer(timeLimit);

            if (int.TryParse(userInput, out int answer) && answer ==
quiz[i].CorrectOption)
            {
                Console.WriteLine("Correct!\n");
                score++;
            }
            else
            {
                Console.WriteLine("Wrong or Time's up! The correct answer was: " +
quiz[i].CorrectOption + "\n");
            }
        }
        Console.WriteLine("Quiz over! Your final score is: " + score + "/10");
    }
    static string ReadInputWithTimer(int seconds)
    {
        string input = "";
        Thread inputThread = new Thread(() => input = Console.ReadLine());
        inputThread.Start();
        bool completed = inputThread.Join(seconds * 1000);
        if (!completed)
        {
            inputThread.Abort();
        }
        return input;
    }
}
```

Output :

```
Question 1: What is the capital of France?
1. Paris
2. London
3. Berlin
4. Madrid
Enter your answer (1-4) within 10 seconds: 1
Correct!
```

```
Question 4: Which gas do plants use for photosynthesis?
1. Oxygen
2. Nitrogen
3. Carbon Dioxide
4. Hydrogen
Enter your answer (1-4) within 10 seconds: 1
Wrong or Time's up! The correct answer was: 3
```

```
Question 10: Which ocean is the largest?
1. Atlantic
2. Indian
3. Arctic
4. Pacific
Enter your answer (1-4) within 10 seconds: 1
Wrong or Time's up! The correct answer was: 4

Quiz over! Your final score is: 7/10
PS C:\Users\AmanDiktiya\Desktop\dotnet\QuizApp>
```

Practical:08

Aim- Develop a database application to store the details of students using ADO.NET

Program.cs:-

```
using System;
using MySql.Data.MySqlClient;

class Program
{
   static string connStr =
"server=localhost;user=root;password=;database=studentdb";

   static void Main()
   {
      Console.WriteLine("1. Add Student");
      Console.WriteLine("2. View Students");
      Console.Write("Enter choice: ");
      int choice = int.Parse(Console.ReadLine());
      if (choice == 1)
         AddStudent();
      else
         ViewStudents();
   }
   static void AddStudent()
   {
      Console.Write("Enter Name: ");
      string name = Console.ReadLine();
      Console.Write("Enter Age: ");
      int age = int.Parse(Console.ReadLine());
      Console.Write("Enter Course: ");
      string course = Console.ReadLine();

      using (MySqlConnection con = new MySqlConnection(connStr))
      {
         con.Open();
         string query = "INSERT INTO students(name,age,course)
VALUES(@name,@age,@course)";
         MySqlCommand cmd = new MySqlCommand(query, con);
         cmd.Parameters.AddWithValue("@name", name);
         cmd.Parameters.AddWithValue("@age", age);
```

```
            cmd.Parameters.AddWithValue("@course", course);
            cmd.ExecuteNonQuery();
            Console.WriteLine("Student Added Successfully!");
        }
    }
    static void ViewStudents()
    {
        using (MySqlConnection con = new MySqlConnection(connStr))
        {
            con.Open();
            string query = "SELECT * FROM students";

            MySqlCommand cmd = new MySqlCommand(query, con);
            MySqlDataReader reader = cmd.ExecuteReader();
            while (reader.Read())
            {
                Console.WriteLine(
                    reader["id"] + " " +
                    reader["name"] + " " +
                    reader["age"] + " " +
                    reader["course"]
                );
            }
        }
    }
}
```

CREATE DATABASE StudentDB;

```
USE StudentDB;

CREATE TABLE Students
(
    StudentID INT PRIMARY KEY IDENTITY(1,1),
    Name VARCHAR(50),
    Course VARCHAR(50),
    Email VARCHAR(50),
    Mobile VARCHAR(15)
);
```

Output :

```
PS C:\Users\AmanDiktiya\Desktop\dotnet\StudentADOApp> dotnet run
1. Add Student
2. View Students
Enter choice: 2
1 Aman Diktiya 20 Information Technology
PS C:\Users\AmanDiktiya\Desktop\dotnet\StudentADOApp> dotnet run
1. Add Student
2. View Students
Enter choice: 1
Enter Name: Shivank
Enter Age: 20
Enter Course: Information Technology
Student Added Successfully!
```

```
PS C:\Users\AmanDiktiya\Desktop\dotnet\StudentADOApp> dotnet run
1. Add Student
2. View Students
Enter choice: 2
1 Aman Diktiya 20 Information Technology
2 Shivank 20 Information Technology
3 Dheeraj  20 Information Technology
4 Abhishek 20 Information Technology
PS C:\Users\AmanDiktiya\Desktop\dotnet\StudentADOApp>
```

Practical:09

Aim- Develop a database application using ADO.NET to insert, modify, update and delete operations.

Program.cs:-

```
using MySql.Data.MySqlClient;
using System;
using System.Data;
using System.Formats.Asn1;
using System.Windows.Forms;
using System.Xml.Linq;

namespace StudentADOApp
{
   public partial class Form1 : Form
   {
      string conStr =
"server=localhost;user=root;password=;database=P8StudentDB;";

      public Form1()
      {
         InitializeComponent();
      }

      // INSERT
      private void btnInsert_Click(object sender, EventArgs e)
      {
         using (MySqlConnection con = new MySqlConnection(conStr))
         {
            con.Open();
            string query = "INSERT INTO Students(Name,RollNo,Class,Marks)
VALUES(@name,@roll,@class,@marks)";
            MySqlCommand cmd = new MySqlCommand(query, con);

            cmd.Parameters.AddWithValue("@name", txtName.Text);
            cmd.Parameters.AddWithValue("@roll", txtRoll.Text);
            cmd.Parameters.AddWithValue("@class", txtClass.Text);
            cmd.Parameters.AddWithValue("@marks", txtMarks.Text);

            cmd.ExecuteNonQuery();
            MessageBox.Show("Record Inserted");
         }
```

```
        }

        // DISPLAY
        private void btnDisplay_Click(object sender, EventArgs e)
        {
            using (MySqlConnection con = new MySqlConnection(conStr))
            {
                con.Open();
                string query = "SELECT * FROM Students";
                MySqlDataAdapter da = new MySqlDataAdapter(query, con);
                DataTable dt = new DataTable();

                da.Fill(dt);
                dataGridView1.DataSource = dt;
            }
        }

        // UPDATE
        private void btnUpdate_Click(object sender, EventArgs e)
        {
            using (MySqlConnection con = new MySqlConnection(conStr))
            {
                con.Open();
                string query = "UPDATE Students SET Name=@name, Class=@class,
Marks=@marks WHERE RollNo=@roll";
                MySqlCommand cmd = new MySqlCommand(query, con);

                cmd.Parameters.AddWithValue("@name", txtName.Text);
                cmd.Parameters.AddWithValue("@roll", txtRoll.Text);
                cmd.Parameters.AddWithValue("@class", txtClass.Text);
                cmd.Parameters.AddWithValue("@marks", txtMarks.Text);

                cmd.ExecuteNonQuery();
                MessageBox.Show("Record Updated");
            }
        }

        // DELETE
        private void btnDelete_Click(object sender, EventArgs e)
        {
            using (MySqlConnection con = new MySqlConnection(conStr))
            {
                con.Open();
                string query = "DELETE FROM Students WHERE RollNo=@roll";
```

```
            MySqlCommand cmd = new MySqlCommand(query, con);

            cmd.Parameters.AddWithValue("@roll", txtRoll.Text);

            cmd.ExecuteNonQuery();
            MessageBox.Show("Record Deleted");
        }
    }
  }
}
```

CREATE DATABASE Practical9;

```
USE Practical9;
CREATE TABLE Students(
    Id INT AUTO_INCREMENT PRIMARY KEY,
    Name VARCHAR(50),
    Age INT,
    Course VARCHAR(50)
);
```

Output :

```
Enter choice: 1
Enter Name: Shivank
Enter Age: 20
Enter Course: .NET
Inserted Successfully
```

Insert

Search Results | Export

Id int(11)	* Name varchar(100)	* Age int(11)	* Course varchar(50)
1	Aman Diktiya	20	AWP
2	Shivank	20	.NET

```
Enter choice: 2
Enter Student ID to update: 3
Enter New Name: Aman
Enter New Age: 21
Enter New Course: .NET
Student updated successfully!
```

Update

Search Results | Export

Id int(11)	* Name varchar(100)	* Age int(11)	* Course varchar(50)
1	Aman	21	.NET
2	Shivank	20	.NET

```
1.Insert
2.Update
3.Delete
4.Display
5.Exit
Enter choice: 3
Enter ID: 3
Deleted Successfully
```

Delete

Search Results | Export

Id int(11)	* Name varchar(100)	* Age int(11)	* Course varchar(50)
4	Shivank	20	.NET

Practical:10

Aim- Develop a application using Data grid to display records.

Program.cs:-

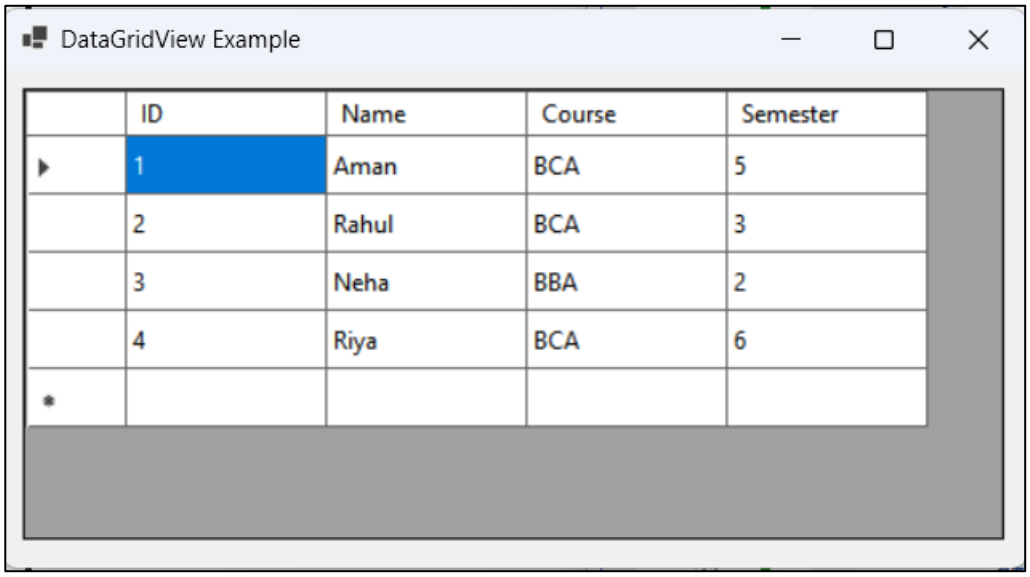
```
using System;
using System.Windows.Forms;
namespace DataGridApp
{
   internal static class Program
   {
      [STAThread]
      static void Main()
      {
         Application.EnableVisualStyles();
         Application.SetCompatibleTextRenderingDefault(false);
         Application.Run(new Form1());
      }
   }
}
```

Form1.cs:-

```
using System;
using System.Data;
using System.Windows.Forms;

namespace DataGridApp
{
   public partial class Form1 : Form
   {
      public Form1()
      {
         InitializeComponent();
      }

      private void Form1_Load(object sender, EventArgs e)
      {
         LoadData();
      }
      private void LoadData()
      {
         DataTable dt = new DataTable();
```

```
            dt.Columns.Add("ID");
            dt.Columns.Add("Name");
            dt.Columns.Add("Course");
            dt.Columns.Add("Semester");

            dt.Rows.Add("1", "Aman", "BCA", "5");
            dt.Rows.Add("2", "Rahul", "BCA", "3");
            dt.Rows.Add("3", "Neha", "BBA", "2");
            dt.Rows.Add("4", "Riya", "BCA", "6");

            dataGridView1.DataSource = dt;
        }
    }
}
```

Output :



Practical:11

Aim- Develop a application using Data grid to add, edit and modify records.

Program.cs:-

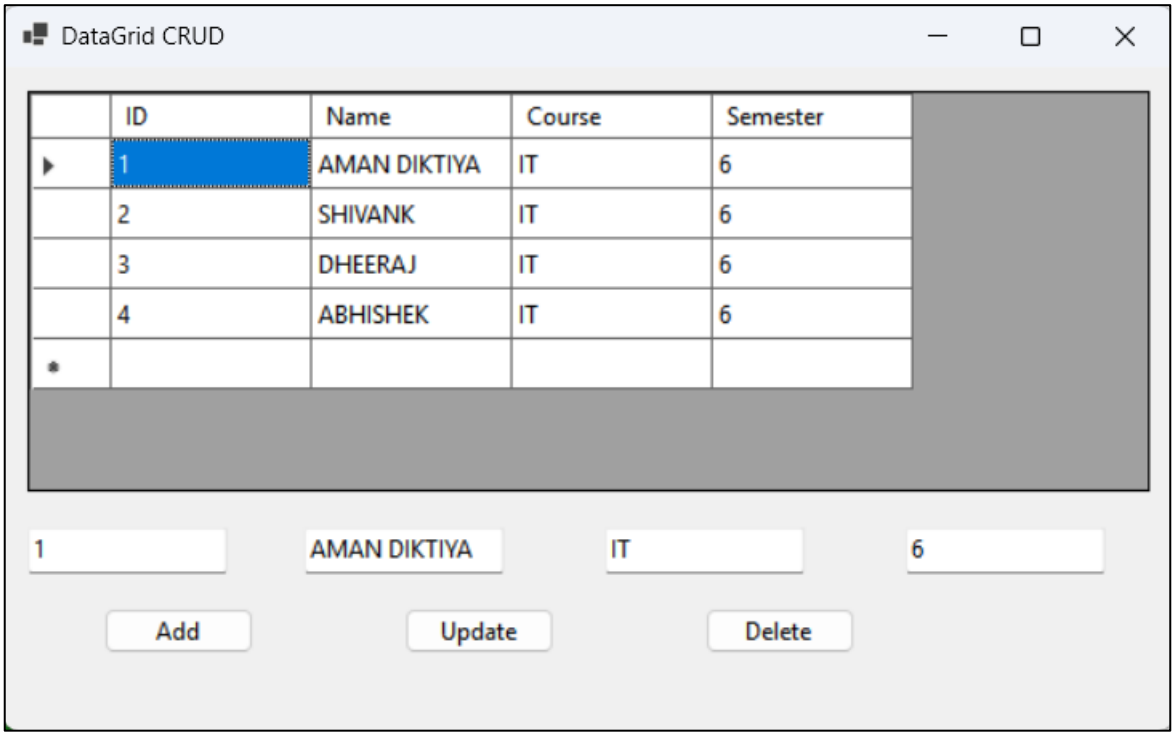
```
using System;
using System.Windows.Forms;

namespace DataGridApp
{
   internal static class Program
   {
      [STAThread]
      static void Main()
      {
         Application.EnableVisualStyles();
         Application.SetCompatibleTextRenderingDefault(false);
         Application.Run(new Form1());
      }
   }
}
```

Form1.cs:-

```
using System;
using System.Data;
using System.Windows.Forms;
namespace DataGridApp
{
   public partial class Form1 : Form
   {
      DataTable dt = new DataTable();

      public Form1()
      {
         InitializeComponent();
      }
      private void Form1_Load(object sender, EventArgs e)
      {
         dt.Columns.Add("ID");
         dt.Columns.Add("Name");
         dt.Columns.Add("Course");
         dt.Columns.Add("Semester");
```

```
            dataGridView1.DataSource = dt;
        }

        private void btnAdd_Click(object sender, EventArgs e)
        {
            dt.Rows.Add(txtId.Text, txtName.Text, txtCourse.Text, txtSemester.Text);
        }
        private void btnUpdate_Click(object sender, EventArgs e)
        {
            if (dataGridView1.CurrentRow != null)
            {
                dataGridView1.CurrentRow.Cells[0].Value = txtId.Text;
                dataGridView1.CurrentRow.Cells[1].Value = txtName.Text;
                dataGridView1.CurrentRow.Cells[2].Value = txtCourse.Text;
                dataGridView1.CurrentRow.Cells[3].Value = txtSemester.Text;
            }
        }
        private void btnDelete_Click(object sender, EventArgs e)
        {
            if (dataGridView1.CurrentRow != null)
            {
                dataGridView1.Rows.Remove(dataGridView1.CurrentRow);
            }
        }

        private void dataGridView1_CellClick(object sender,
DataGridViewCellEventArgs e)
        {
            if (e.RowIndex >= 0)
            {
                DataGridViewRow row = dataGridView1.Rows[e.RowIndex];

                txtId.Text = row.Cells[0].Value.ToString();
                txtName.Text = row.Cells[1].Value.ToString();
                txtCourse.Text = row.Cells[2].Value.ToString();
                txtSemester.Text = row.Cells[3].Value.ToString();
            }
        }
    }
}
```

Output :


DataGrid CRUD
ID
Name
Course
Semester
1
AMAN DIKTIYA
IT
6
2
SHIVANK
IT
6
3
DHEERAJ
IT
6
4
ABHISHEK
IT
6
1
AMAN DIKTIYA
IT
6
Add
Update
Delete

Practical:12

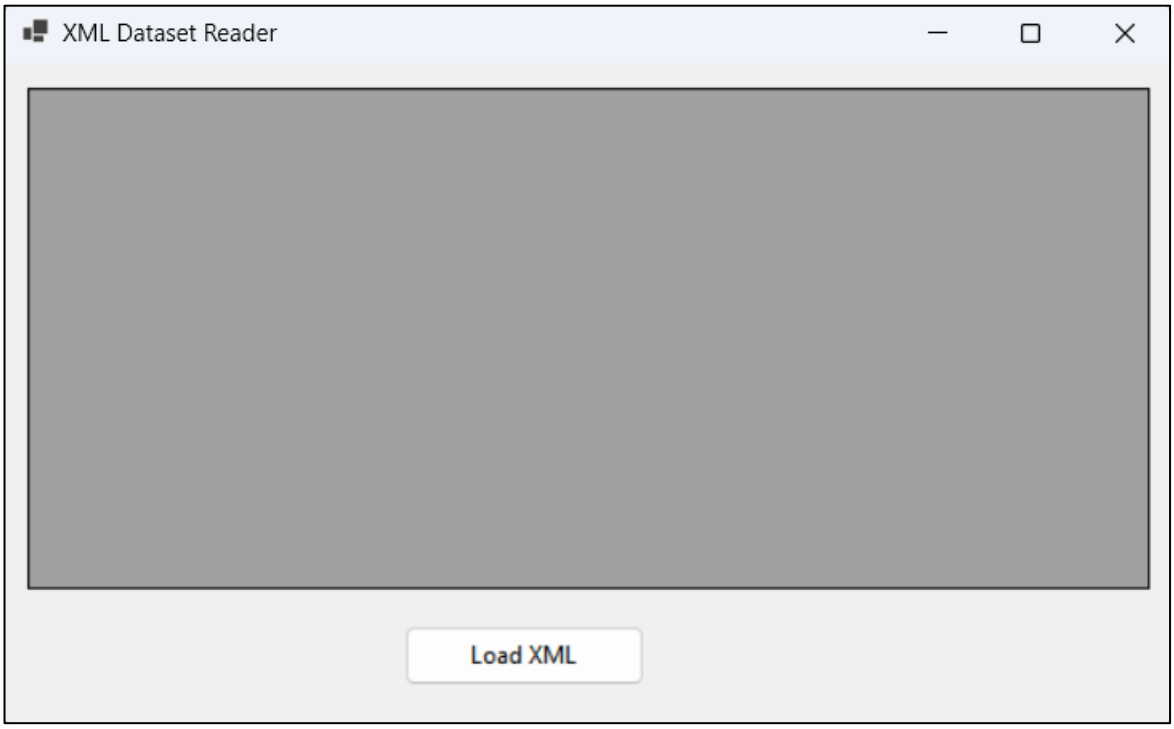
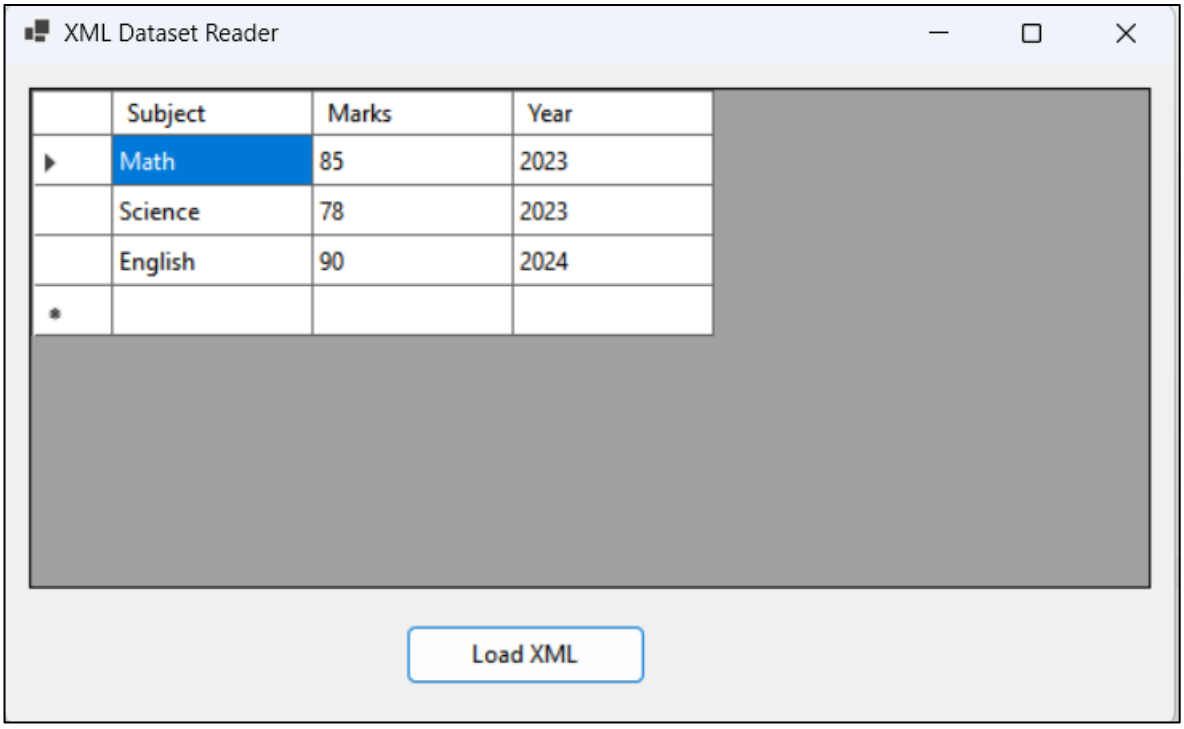
Aim- Develop a Window application to read an XML document containing subject, mark scored, year of passing into a Dataset.

Program.cs:-

```
using System;
using System.Windows.Forms;

namespace XMLDatasetApp
{
    internal static class Program
    {
        [STAThread]
        static void Main()
        {
            Application.EnableVisualStyles();
            Application.SetCompatibleTextRenderingDefault(false);
            Application.Run(new Form1());
        }
    }
}
```

Form1.cs:-

```
using System;
using System.Data;
using System.Windows.Forms;

namespace XMLDatasetApp
{
    public partial class Form1 : Form
    {
        public Form1()
        {
            InitializeComponent();
        }

        private void btnLoad_Click(object sender, EventArgs e)
        {
            DataSet ds = new DataSet();
            ds.ReadXml("marks.xml");
```

```
            dataGridView1.DataSource = ds.Tables[0];
        }
    }
}
```

Output :